

Generalized Braille: From Tactile Code to Inspectable Semantic Vector Space

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Abstract

We formalize Braille as a family of structured codes over signed, extensible, finite-dimensional vector spaces. Standard 6-dot Braille is the instance $\mathcal{B}_6 = \{0, 1\}^6$; 8-dot computer Braille is $\mathcal{B}_8 = \{0, 1\}^8$. We generalize to n -dot Braille over the ternary alphabet $\{-1, 0, +1\}^n$, which introduces explicit negation, inhibition, and algebraic structure absent from binary encodings. We show that composing ternary cells into sequences yields a representational system with no fixed semantic-complexity ceiling, and that the limiting factor shifts from vocabulary size to the quality of the learned composition law. We further extend to unbounded-dot and infinity-dot Braille, connecting the framework to sparse feature spaces and neural latent representations. We describe a concrete implementation as a pre-training objective for convolutional style-transfer networks, and present curriculum experiments spanning Nemeth mathematical Braille, signed-dot patterns, and n -dot cells for $n \in \{6, 8, 10, 12\}$.

1 Introduction

Braille is conventionally understood as a tactile writing system: a fixed grid of raised dots encoding characters from a finite alphabet. Standard Braille uses a 2×3 grid of 6 dots, yielding $2^6 = 64$ possible cell patterns. Computer Braille extends this to a 2×4 grid of 8 dots, yielding $2^8 = 256$ patterns.

In both cases, a cell is a binary vector:

$$\mathbf{b} = (b_1, \dots, b_n), \quad b_i \in \{0, 1\}, \quad (1)$$

and the set of all cells is:

$$|\mathcal{B}_n| = 2^n. \quad (2)$$

We observe that this is an instance of a much richer family of structured codes. The natural generalization replaces the binary alphabet with a signed ternary alphabet, extends the dimensionality to arbitrary n , and considers composition of cells into sequences. The result is a framework that bridges tactile encoding, symbolic representation, and neural feature spaces.

2 The n -dot Braille Family

Definition 1 (n -dot binary Braille). *An n -dot binary Braille cell is a vector $\mathbf{b} \in \{0, 1\}^n$, arranged in a $2 \times \lceil n/2 \rceil$ spatial grid. The cell space has cardinality 2^n .*

The standard instances form a nested hierarchy:

$$\mathcal{B}_6 \subset \mathcal{B}_8 \subset \mathcal{B}_{10} \subset \dots \quad (3)$$

assuming the added dimensions preserve the semantics of earlier cells (i.e., dots 1–6 retain their standard meanings).

System	n	Grid	$ \mathcal{B}_n $
Standard Braille	6	2×3	64
Computer Braille	8	2×4	256
Extended Braille	10	2×5	1,024
12-dot Braille	12	2×6	4,096

Table 1: Binary n -dot Braille instances.

The spatial arrangement matters. A cell is not merely a bit string; it is a *spatially grounded* bit string. The 2-column structure assigns each dot a position (c, r) where $c \in \{0, 1\}$ (left/right) and $r \in \{0, \dots, \lceil n/2 \rceil - 1\}$ (row). This spatial grounding is what distinguishes Braille from an arbitrary binary code and makes it learnable by spatial perception systems.

3 Signed Braille: The Ternary Extension

Definition 2 (Signed n -dot Braille). *A signed Braille cell is a vector $\mathbf{b} \in \{-1, 0, +1\}^n$, where:*

$$b_i = +1 \quad (\text{asserted: the feature is present}), \quad (4)$$

$$b_i = 0 \quad (\text{unspecified: no statement about feature } i), \quad (5)$$

$$b_i = -1 \quad (\text{denied: the feature is explicitly absent}). \quad (6)$$

The cell space has cardinality 3^n . For 8-dot signed Braille: $3^8 = 6,561$ patterns. For 10-dot: $3^{10} = 59,049$.

Remark 1. *Binary Braille cannot distinguish “not present” from “explicitly absent.” The value $b_i = 0$ in $\{0, 1\}^n$ conflates two distinct semantic states. Signed Braille resolves this conflation.*

3.1 Algebraic Properties

Signed cells admit natural algebraic operations.

Negation. For $\mathbf{x} = (x_1, \dots, x_n)$:

$$-\mathbf{x} = (-x_1, \dots, -x_n). \quad (7)$$

This corresponds to conceptual inversion: every asserted feature becomes denied, and vice versa.

Cancellation.

$$(+d_i) + (-d_i) = 0. \quad (8)$$

A denied dot cancels an asserted dot, yielding the unspecified state.

Inner product.

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n x_i y_i. \quad (9)$$

This measures alignment versus opposition:

$$x_i y_i = \begin{cases} +1 & \text{agreement (both asserted or both denied),} \\ 0 & \text{irrelevant (at least one unspecified),} \\ -1 & \text{contradiction (one asserted, one denied).} \end{cases} \quad (10)$$

Signed similarity. Two cells agree when their signs align and conflict when they oppose. This permits distance metrics, clustering, and analogy operations directly on the cell representation.

4 Compositional Expressivity

A single n -dot ternary cell carries:

$$\log_2(3^n) = n \log_2 3 \approx 1.585n \text{ bits.} \quad (11)$$

For 10-dot ternary Braille, this is approximately 15.85 bits per cell. Substantial, but finite. The real expressivity arises from *composition*. A sequence of L cells:

$$\mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_L), \quad (12)$$

lives in the product space:

$$|\mathcal{X}_L| = (3^n)^L = 3^{nL}. \quad (13)$$

At only 32 cells of 10-dot ternary Braille:

$$3^{320} \approx 10^{153}. \quad (14)$$

But state counting misses the point. English has fewer than 100 primitive characters, yet:

$$\text{small alphabet} + \text{composition} \rightarrow \text{Shakespeare, mathematics, Python.} \quad (15)$$

Binary computers are even more extreme: $\{0, 1\}$ suffices for essentially all digitally representable knowledge.

Theorem 1 (Expressivity hierarchy). *Ternary n -dot Braille with composition admits three levels of expressivity:*

1. **Atomic:** 3^n meanings (finite, fixed).
2. **Linguistic:** $(\{-1, 0, +1\}^n)^*$ — infinitely many finite expressions encoding sentences, trees, graphs, proofs, and programs.
3. **Computational:** With read, write, branch, bind, and recursion operations plus unbounded working memory, the system is Turing-complete.

Therefore:

$$\boxed{\{-1, 0, +1\}^n + \text{composition} \rightarrow \text{arbitrary finite symbolic structures}} \quad (16)$$

and:

$$\boxed{\{-1, 0, +1\}^n + \text{composition} + \text{recursion} + \text{memory} \rightarrow \text{general computation}} \quad (17)$$

5 Factored Representations

The crucial design principle: meaning should *factor through the dots*.

An arbitrary lookup table assigns:

$$\mathbf{b} \mapsto \text{symbol}. \quad (18)$$

A semantic Braille system assigns:

$$b_i \mapsto \text{primitive semantic feature} \quad (19)$$

and therefore:

$$\mathbf{b} \mapsto \text{composition of active features}. \quad (20)$$

Increasing n then adds *semantic axes* rather than merely vocabulary capacity.

The cell-level embedding becomes:

$$\phi(\mathbf{b}) = \sum_{i=1}^n b_i \mathbf{d}_i + \sum_{i < j} b_i b_j \mathbf{d}_{ij} + \dots \quad (21)$$

where $\mathbf{d}_i$ represents a primitive dot meaning and $\mathbf{d}_{ij}$ represents pairwise interactions.

Sequence-level composition adds another layer:

$$\Phi(\mathbf{x}_1, \dots, \mathbf{x}_L) = F(\phi(\mathbf{x}_1), \dots, \phi(\mathbf{x}_L)). \quad (22)$$

The full hierarchy is:

$$\text{trit} \rightarrow \text{dot interactions} \rightarrow \text{cell} \rightarrow \text{cell interactions} \rightarrow \text{structure} \rightarrow \text{program/meaning}. \quad (23)$$

Remark 2. *Past some n , exhaustive learning of all 3^n cells becomes impossible. The system must learn the factorization of dots (Equation 21), not every possible cell independently. This is the regime where factored representations become essential.*

6 Extensions Beyond Ternary

The ternary alphabet is not the endpoint.

Definition 3 (Generalized dot algebra). *A dot may take values in an arbitrary domain D_i :*

$$b_i \in \mathbb{Z} \quad (\text{integer: count, multiplicity}), \quad (24)$$

$$b_i \in \mathbb{R} \quad (\text{real: activation, confidence}), \quad (25)$$

$$b_i \in \mathbb{C} \quad (\text{complex: magnitude + phase}), \quad (26)$$

$$b_i \in [0, 1] \quad (\text{probabilistic: belief}), \quad (27)$$

$$b_i \in [l_i, u_i] \quad (\text{interval: uncertainty bounds}). \quad (28)$$

The most coherent hierarchy is:

$$\text{ordinary Braille} = \{0, 1\}^n, \quad (29)$$

$$\text{signed Braille} = \{-1, 0, +1\}^n, \quad (30)$$

$$\text{weighted Braille} = \mathbb{R}^n, \quad (31)$$

$$\text{infinity-dot Braille} = \mathbb{R}^{\mathbb{N}} \text{ with sparse or computable support}. \quad (32)$$

This converges toward a learned latent space, but with a critical difference: **the dimensions remain externally addressable and potentially inspectable**.

A neural vector is usually $\mathbf{z} \in \mathbb{R}^d$ with opaque coordinates. Generalized Braille is $\mathbf{b} \in \mathbb{R}^d$ with spatially or semantically grounded coordinates.

Generalized Braille = an inspectable, extensible, signed semantic vector space

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7 Infinity-dot Braille

7.1 Unbounded Braille

The cell may contain any finite number of dimensions, with no fixed maximum:

$$\mathbf{b} \in \bigcup_{n=1}^{\infty} \{0, 1\}^n. \quad (34)$$

This is operationally sensible. A representation can expand when it needs new semantic dimensions. It resembles a dynamically extensible schema:

$$\mathbf{b} = (b_1, \dots, b_n), \quad n \text{ chosen per expression.} \quad (35)$$

This is the useful form of “infinity-dot Braille.”

7.2 Actually Infinite Braille

A cell is an infinite binary sequence:

$$\mathbf{b} \in \{0, 1\}^{\mathbb{N}}. \quad (36)$$

The complete space is uncountable:

$$\left| \{0, 1\}^{\mathbb{N}} \right| = 2^{\aleph_0}. \quad (37)$$

But computation can only access finite prefixes. Practical infinity-dot Braille must impose one of:

- Finite support: $\sum_i b_i < \infty$.
- Computable sequence: $b_i = f(i)$ for computable f .
- Lazy representation: $b_{1:k}$ generated on demand.

A sparse infinite cell is especially natural:

$$\mathbf{b} : \mathbb{N} \rightarrow \{0, 1\}, \quad |\{i : b_i = 1\}| < \infty. \quad (38)$$

This gives an unlimited namespace while keeping every actual expression finite:

$$\mathbf{b} \equiv \{i_1, i_2, \dots, i_k\}. \quad (39)$$

Infinity-dot Braille is a semantic coordinate system.

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8 Implementation: Pre-training Objective

We implement generalized Braille as a pre-training objective for convolutional style-transfer networks.

8.1 Synthetic Data Generation

For n -dot signed Braille with n configurable:

1. Generate all 3^n (or 2^n) possible cell patterns.
2. Render cells as images on a $2 \times \lceil n/2 \rceil$ spatial grid.
3. Asserted dots (+1) are rendered as filled circles.
4. Denied dots (−1) are rendered as X marks.
5. Absent dots (0) produce no mark.
6. Add position jitter, size variation, and pressure simulation to approximate hand-drawn appearance.

8.2 Network Architecture

The pre-training target is a convolutional encoder-decoder (TransformerNet) trained with perceptual loss via VGG-16 features. The hypothesis: teaching the network that spatial dot patterns carry structured, compositionally meaningful information produces better spatial preservation during style transfer.

8.3 Training Protocol

1. **Stage 1 (Braille):** Pre-train on synthetic signed n -dot Braille images. Content weight is doubled to emphasize spatial fidelity.
2. **Stage 2 (Style transfer):** Fine-tune on natural images (COCO) with standard style loss.

8.4 Curriculum

The drawing curriculum includes:

- 8-dot lowercase and uppercase alphabet (dot 7 as capital indicator).
- Nemeth mathematical Braille: integrals, summations, Greek letters, the quadratic formula, Euler’s identity, the Pythagorean theorem, and the Basel problem.
- Generalized n -dot hierarchy: $\mathcal{B}_6 \subset \mathcal{B}_8 \subset \mathcal{B}_{10}$ rendered side-by-side.
- Negative-dot demonstrations: asserted, denied, and absent in the same cell.
- 12-dot cells (2×6 grid, 4,096 binary patterns).
- Algebraic cancellation: $(+d) + (-d) = 0$.

9 Conclusion

We have shown that Braille, properly generalized, is not a fixed alphabet but a *semantic vector space* with the following properties:

1. **Spatial grounding:** Dot positions are physically meaningful, not arbitrary indices.
2. **Signed semantics:** The ternary alphabet $\{-1, 0, +1\}$ distinguishes assertion, absence, and denial.

3. **Compositionality:** Cell sequences can encode arbitrary finite structures.
4. **Extensibility:** Adding dimensions adds semantic axes, not just capacity.
5. **Inspectability:** Unlike opaque neural embeddings, dot coordinates are externally addressable.
6. **Algebraic structure:** Negation, cancellation, and inner products operate directly on the representation.

The key results:

Finite n -dot Braille	is a vocabulary
Unbounded-dot Braille	is a growing ontology
Infinity-dot Braille	is a semantic basis
Negative-dot Braille	adds contradiction and inhibition

A fixed ternary n -dot cell is finite. But an arbitrarily composable ternary Braille language has *no fixed semantic-complexity ceiling*. The limiting factor is not the number of dots. It is the quality of the learned composition law.

$$\boxed{\text{Generalized Braille} = \text{an inspectable, extensible, signed semantic vector space.}} \quad (41)$$